

MODULE-1

Session 3

(TASKS)

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans:

- **HTTP – Hyper-Text Transfer Protocol: -**

HTTP is a communication protocol used for transferring web pages, images, videos, and other data between a web browser and a web server. It forms the foundation of data communication on the World Wide Web.

- **HTTPS – HyperText Transfer Protocol Secure: -**

HTTPS is the secure version of HTTP that uses encryption to protect data exchanged between users and websites. It helps ensure confidentiality, integrity, and security of information such as passwords and online transactions.

- **FTP – File Transfer Protocol: -**

FTP is a standard network protocol used to transfer files from one computer to another over the internet or a local network. It is commonly used by web developers to upload website files to web servers.

- **SMTP – Simple Mail Transfer Protocol: -**

SMTP is a protocol used for sending, relaying, and forwarding email messages between email clients and mail servers. It ensures that emails are delivered to the intended recipients.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans:

- **IPv4 (Internet Protocol Version 4): -**

uses a 32-bit addressing scheme and represents addresses in decimal format, such as **192.168.1.1**. It can support approximately 4.3 billion unique IP addresses, which are becoming insufficient due to the rapid growth of internet-connected devices.

- **IPv6 (Internet Protocol Version 6): -**

uses a 128-bit addressing scheme and represents addresses in hexadecimal format, such as **2001:0db8:85a3::8a2e:0370:7334**. It provides an extremely large number of unique addresses, improved security features, and more efficient network routing compared to IPv4.

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3. Run nslookup google.com on your system and write down the IP address you get.

Ans:

- **I cannot run commands on your computer, but you can do it yourself by following these steps: -**

1. Open Command Prompt on Windows.

2. Type the following command: -

nslookup google.com

3. Press Enter.

You may receive an output similar to: -

Name: google.com
Address: 142.250.193.78

Note: - The IP address may vary depending on your location, internet service provider, and Google's load-balancing system. Therefore, different users may obtain different IP addresses for the same domain.

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Ans:

- **200 – OK: -**

This status code indicates that the request was successfully processed by the server and the requested resource has been returned to the user without any issues.

- **301 – Moved Permanently: -**

This status code indicates that the requested webpage has been permanently moved to a new URL. Browsers automatically redirect users to the new location.

- **404 – Not Found: -**

This status code indicates that the requested webpage or resource does not exist on the server. It usually occurs when a page has been deleted or the URL has been entered incorrectly.

- **500 – Internal Server Error: -**

This status code indicates that the server encountered an unexpected problem while processing the request. The error is generally caused by issues within the server application.

- **403 – Forbidden: -**

This status code indicates that the server understands the request but refuses to provide access because the user does not have the necessary permissions.

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5. Explain DNS hierarchy (root, TLD, domain, subdomain) with **www.google.com** as an example.

Ans:

- The **Domain Name System (DNS)** is a hierarchical naming system that converts human-readable domain names into IP addresses. DNS makes it possible for users to access websites using easy-to-remember names instead of numerical IP addresses.

Consider the website **www.google.com**.

1. Root Domain (.): -

The root domain is the highest level in the DNS hierarchy and is represented by a dot (.) at the end of a domain name. Although it is usually hidden, all domain names ultimately originate from the root domain.

2. Top-Level Domain (TLD) – .com: -

The TLD is the highest visible part of a domain name. In **www.google.com**, **.com** is the TLD. Examples of other TLDs include **.org**, **.edu**, **.gov**, and **.in**.

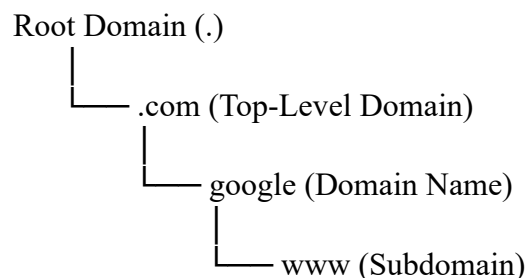
3. Domain Name – google: -

The domain name is the unique name registered by an organization or individual. In this example, **google** identifies Google's services and websites on the internet.

4. Subdomain – www: -

A subdomain is an extension added before the main domain name to organize services or sections of a website. In **www.google.com**, **www** is the subdomain that traditionally represents the web service.

- **DNS Hierarchy Structure: -**



- **How DNS Works: -**

When a user type **www.google.com** into a browser, the DNS system first checks the root servers, then the **.com** servers, followed by Google's domain servers. These servers provide the corresponding IP address, allowing the browser to connect to Google's web server and display the website.

Thus, the DNS hierarchy helps organize the internet and enables users to access websites quickly and efficiently using domain names instead of complex numerical addresses.